Skills Summary

Quadrilaterals: Which Way Does the Implication Go?

Use this reference while completing your worksheet or when studying.

The family tree runs one way. square ⇒ rhombus ⇒ parallelogram ⇒ quadrilateral, and square ⇒ rectangle ⇒ parallelogram. Reading *down* the tree is always safe; reading *up* it needs an extra property, and that extra property is what every problem on the worksheet is testing.

1. What a property is enough for (necessary-vs-sufficient)

Upgrades that work (each is a genuine theorem):

parallelogram + one right angle ⇒ rectangle

parallelogram + congruent diagonals ⇒ rectangle

parallelogram + two congruent adjacent sides ⇒ rhombus

parallelogram + perpendicular diagonals ⇒ rhombus

rhombus + one right angle ⇒ square

rectangle + congruent adjacent sides ⇒ square

Not enough: congruent sides alone never gives right angles, and congruent diagonals alone never gives congruent sides.

Example: A rhombus has consecutive angles $(3x+20)^\circ$ and $(2x+10)^\circ$. Find x, then decide whether it must be a square.
Step 1. A rhombus is a parallelogram, so consecutive angles are supplementary — that is the equation to write, and the resulting angles are what decide "square".
Step 2. $(3x+20)+(2x+10)=180$, so $5x+30=180$. ⇒ $\boldsymbol{x = 30}$, giving angles 110° and 70°: not right angles, so a rhombus and *not* a square.

Try it: A rhombus has consecutive angles $(4x)^\circ$ and $(5x)^\circ$. Find x.

check: $\boldsymbol{x = 20}$

2. Testing a claim with side lengths (coordinate-classification)

Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Two *adjacent* sides of unequal length kill "rhombus" and "square" instantly — one counterexample side pair is all it takes. Perpendicular diagonals give a rhombus only when they also bisect each other (equal midpoints); without that you have a kite.

Example: $A(0,5)$, $B(2,0)$, $C(0,-3)$, $D(-2,0)$: the diagonals lie on the axes, so they are perpendicular. Must $ABCD$ be a rhombus?
Step 1. Perpendicular diagonals are not sufficient on their own, so test two adjacent sides rather than trusting the claim.
Step 2. $AB = \sqrt{2^2+5^2} = \sqrt{29}$, while $BC = \sqrt{2^2+3^2} = \sqrt{13} \approx 3.61$. ⇒ $AB \approx \mathbf{5.39} \neq BC$, so not a rhombus — it is a kite.

Try it: $A(0,7)$, $B(2,0)$, $C(0,-4)$, $D(-2,0)$. Find AB to two decimal places (and see that BC differs).

check: **5.39** becomes **7.28**

3. The diagonal test for the most specific name (most-specific-name)

Rule: once all four sides are known congruent (a rhombus), compare the *diagonals*:

diagonals congruent ⇒ square   diagonals not congruent ⇒ rhombus only.

The same comparison run on a rectangle tells you nothing new — a rectangle always has congruent diagonals. Order matters: sides first, then diagonals.

Example: $A(0,0)$, $B(3,1)$, $C(4,-2)$, $D(1,-3)$ has four congruent sides. Square or rhombus?
Step 1. All sides congruent already gives a rhombus, so the deciding test is whether the diagonals are congruent.
Step 2. $AC = \sqrt{4^2+2^2} = \sqrt{20}$ and $BD = \sqrt{2^2+4^2} = \sqrt{20}$ — equal. ⇒ $AC = \mathbf{4.47} = BD$, so $ABCD$ is a square.

Try it: The rhombus $E(0,0)$, $F(4,2)$, $G(8,0)$, $H(4,-2)$. Find diagonal EG, compare it with $FH = 4$, and name the figure.

check: $EG = \mathbf{8}$, so rhombus only

4. Killing a converse with a counterexample (counterexample-test)

Rule: to show an implication does *not* run backwards, produce one figure with the property but without the name. The classic pairs:

perpendicular diagonals without a rhombus → **kite**   congruent diagonals without a rectangle → **isosceles trapezoid**

The missing hypothesis in both is "and it is a parallelogram".

Example: Is "congruent diagonals ⇒ rectangle" always true? Test $T(0,0)$, $U(6,0)$, $V(5,3)$, $W(1,3)$.
Step 1. To refute a converse I only need one figure with congruent diagonals that is not a rectangle, so compute both diagonals of this trapezoid.
Step 2. $TV = \sqrt{5^2+3^2} = \sqrt{34}$ and $UW = \sqrt{5^2+3^2} = \sqrt{34}$: congruent, yet TU and WV are the only parallel pair and no angle is right. ⇒ $TV = \mathbf{5.83} = UW$ in a shape that is *not* a rectangle.

Try it: $T(0,0)$, $U(8,0)$, $V(6,4)$, $W(2,4)$. Find TV (then check UW matches).

check: **7.21**